



Internet of Things and Wireless Sensor Networks

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Where It All Started...

The most profound technologies are those that disappear.
They weave themselves into the fabric of everyday life until they are indistinguishable from it.

(Mark Weiser, 1991)

Calm technology

Ubiquitous computing

Pervasive computing

Ambient intelligence

Smart spaces

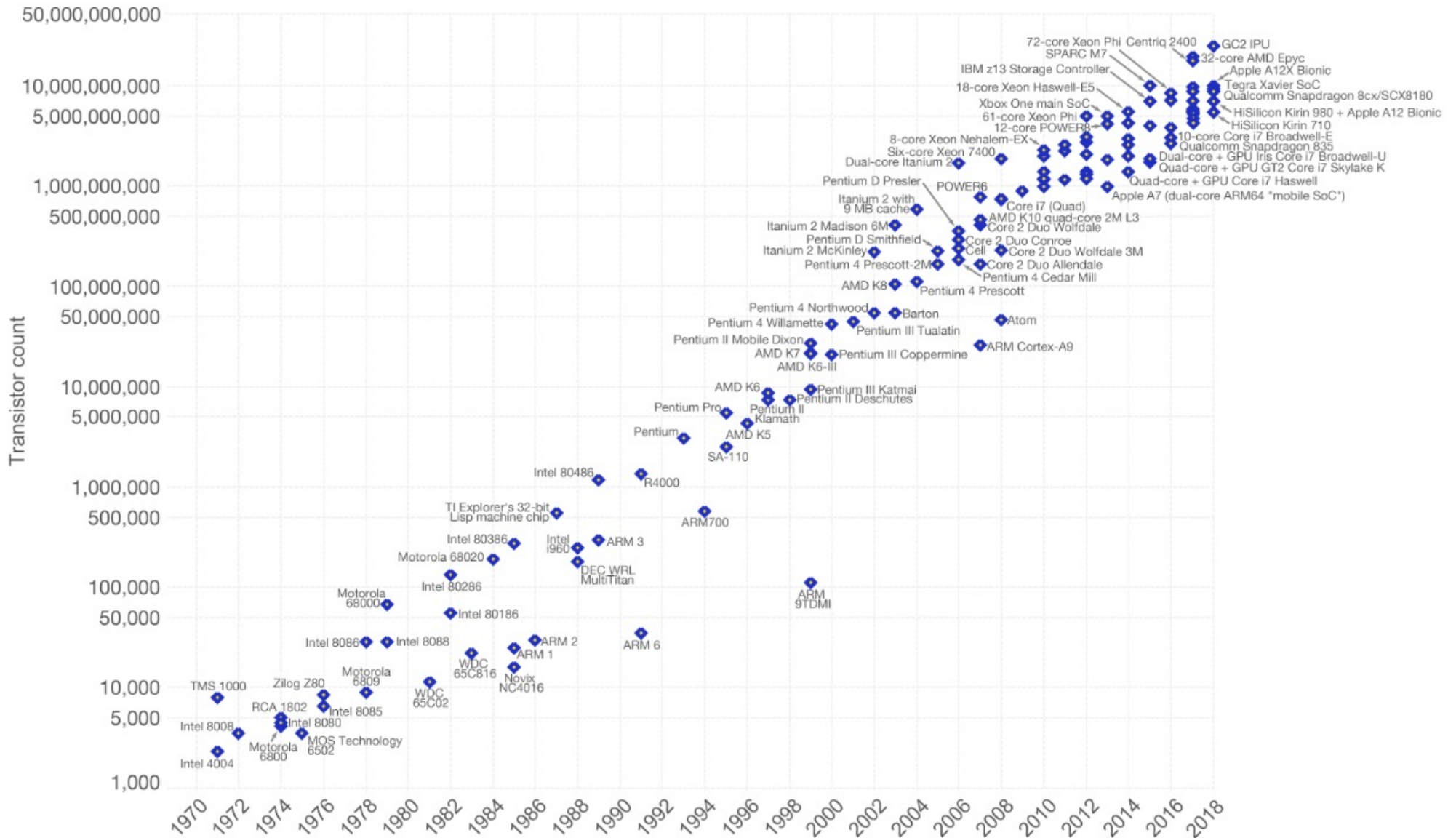
...



... 1991



Miniaturization: Moore's Law



Data source: Wikipedia (https://en.wikipedia.org/wiki/Transistor_count)
The data visualization is available at [OurWorldinData.org](https://ourworldindata.org). There you find more visualizations and research on this topic.

Licensed under [CC-BY-SA](#) by the author Max Roser.

The beginning: one computer, many people



The PC: one computer per person



The future: many computers per person

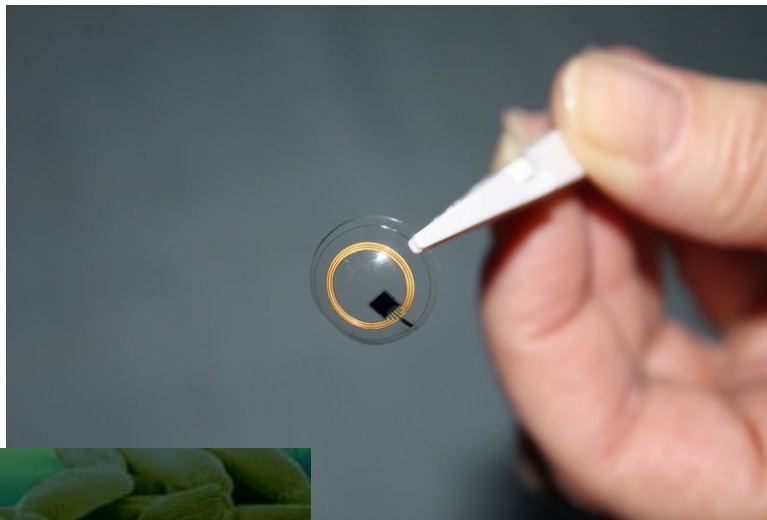


[...] Next comes ubiquitous computing, or the age of calm technology, when technology recedes into the background of our lives.

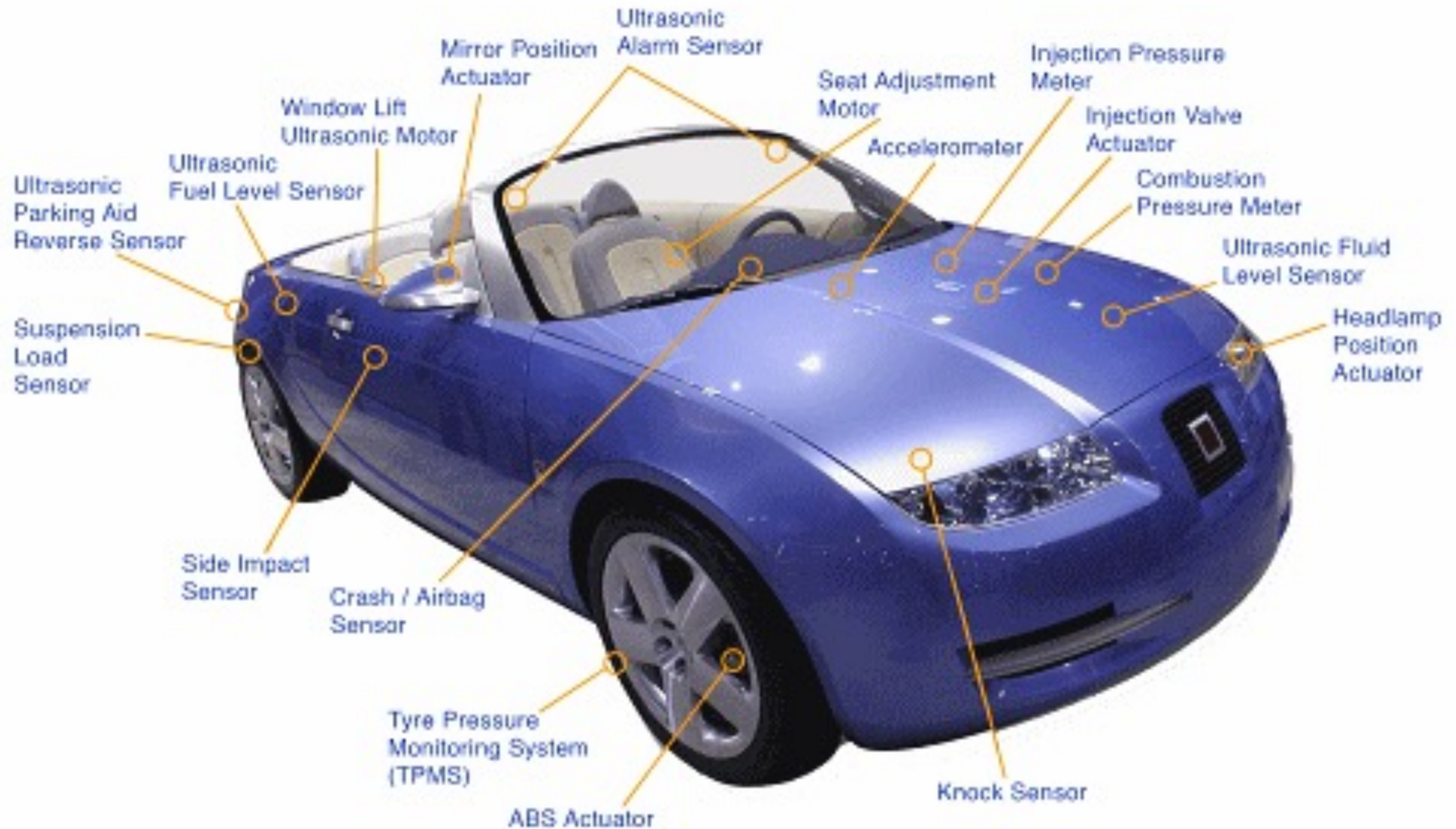


It's not just the computer getting smaller...

- MEMS technology (Micro Electro-Mechanical Systems) for sensors, actuators, ...



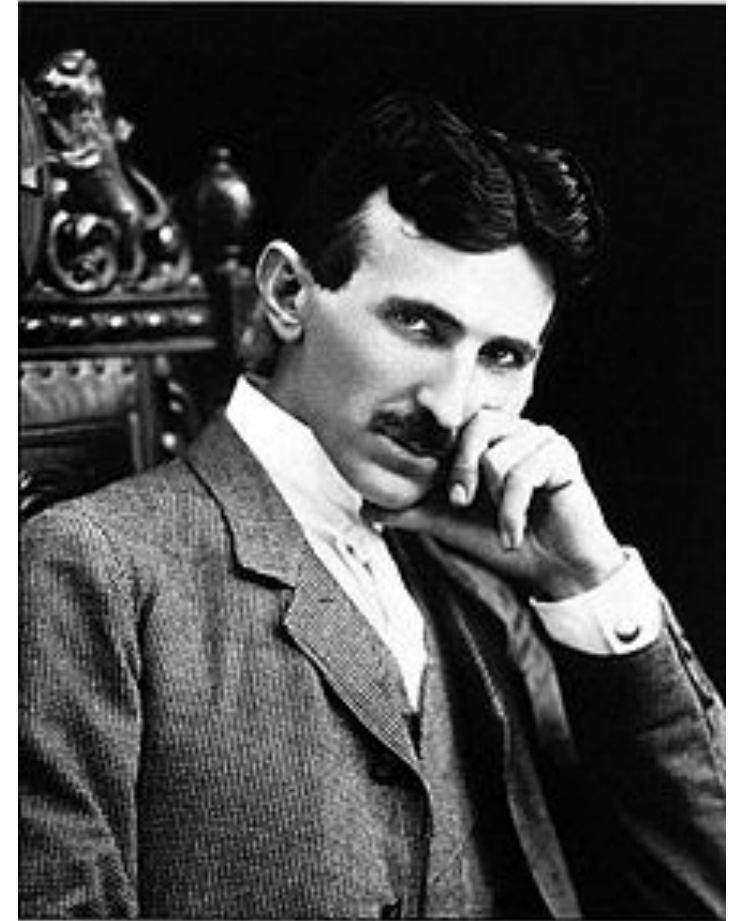
Sensors everywhere!



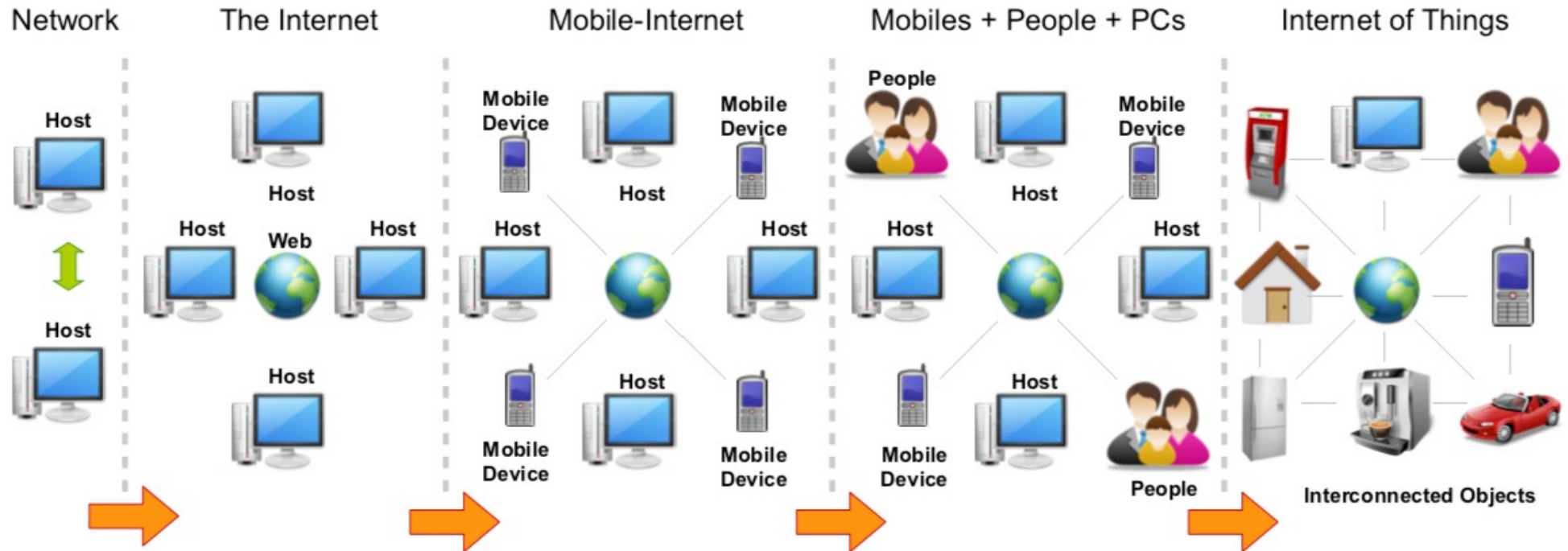
Another visionary

When wireless is perfectly applied the whole earth will be converted into a huge brain [...] and the instruments through which we shall be able to do this will be amazingly simple compared with our present telephone. A man will be able to carry one in his vest pocket.

(Nikola Tesla, 1926)



The Evolution of the Internet

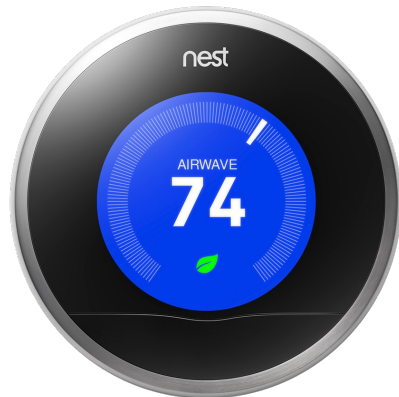
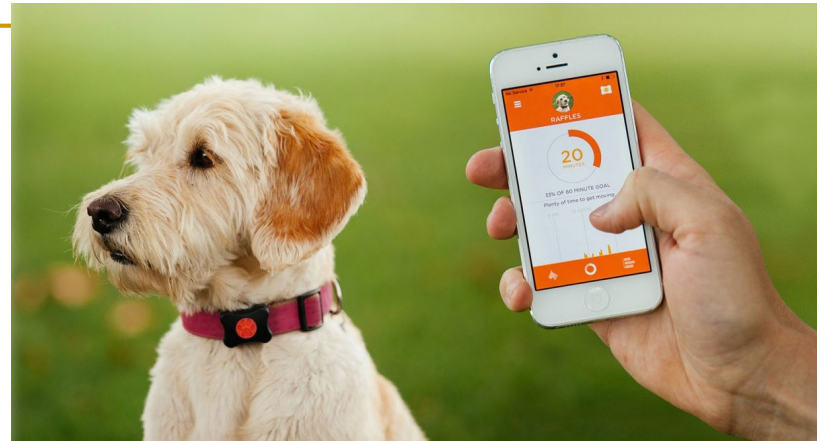


Internet of Things

- A set of interconnected computing devices that:
 - ❑ Autonomously coordinate
 - ❑ Acquire data (sensors) and operate (actuators) in the physical world
 - ❑ Are seamlessly integrated with information systems and the Internet



Some Examples...

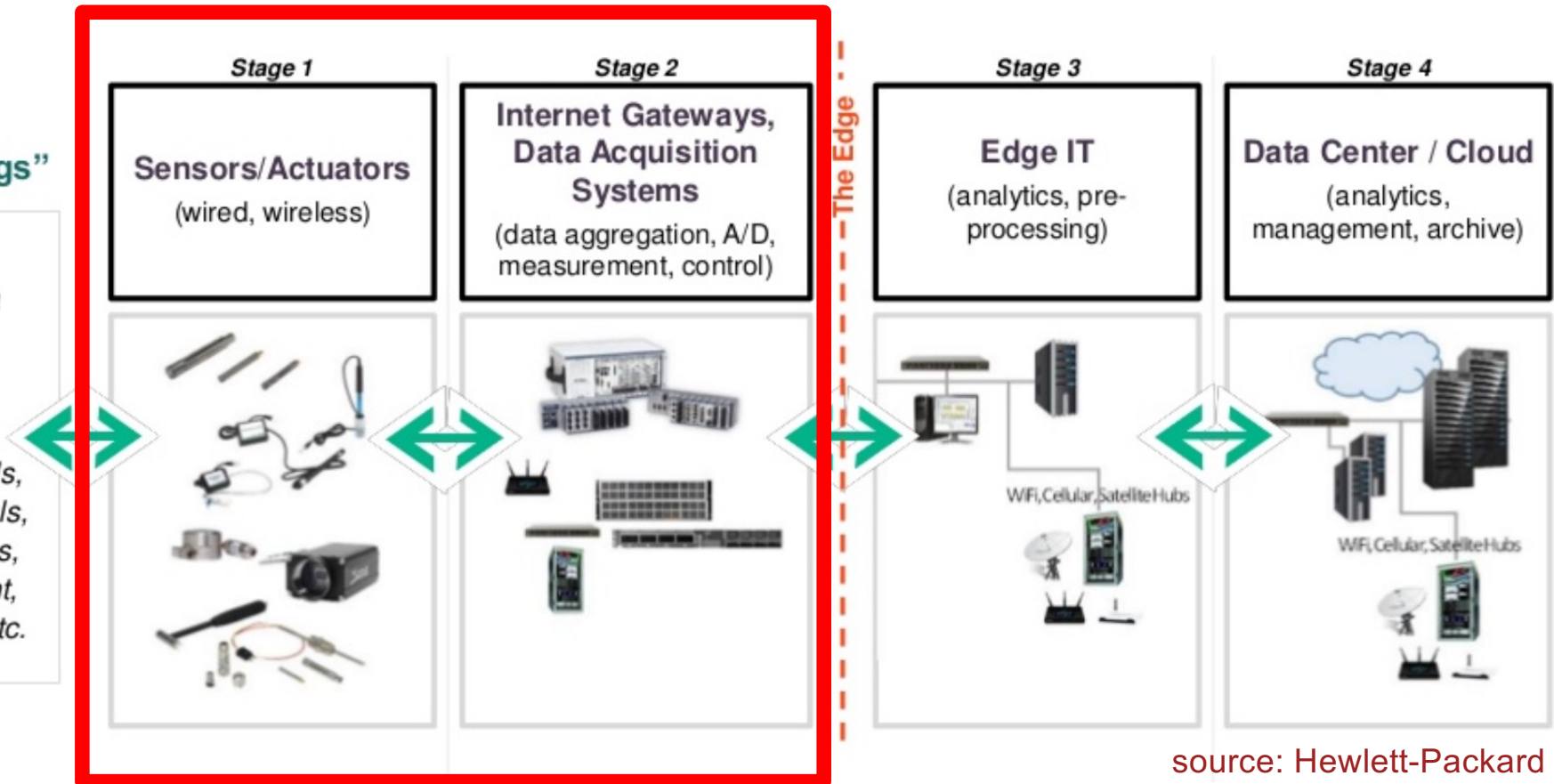


Architecture(s) of IoT

The “Things”

*Primarily
analog data
sources*

*Devices,
machines,
people, tools,
cars, animals,
clothes, toys,
environment,
buildings, etc.*



source: Hewlett-Packard

An example: The “smart cities”



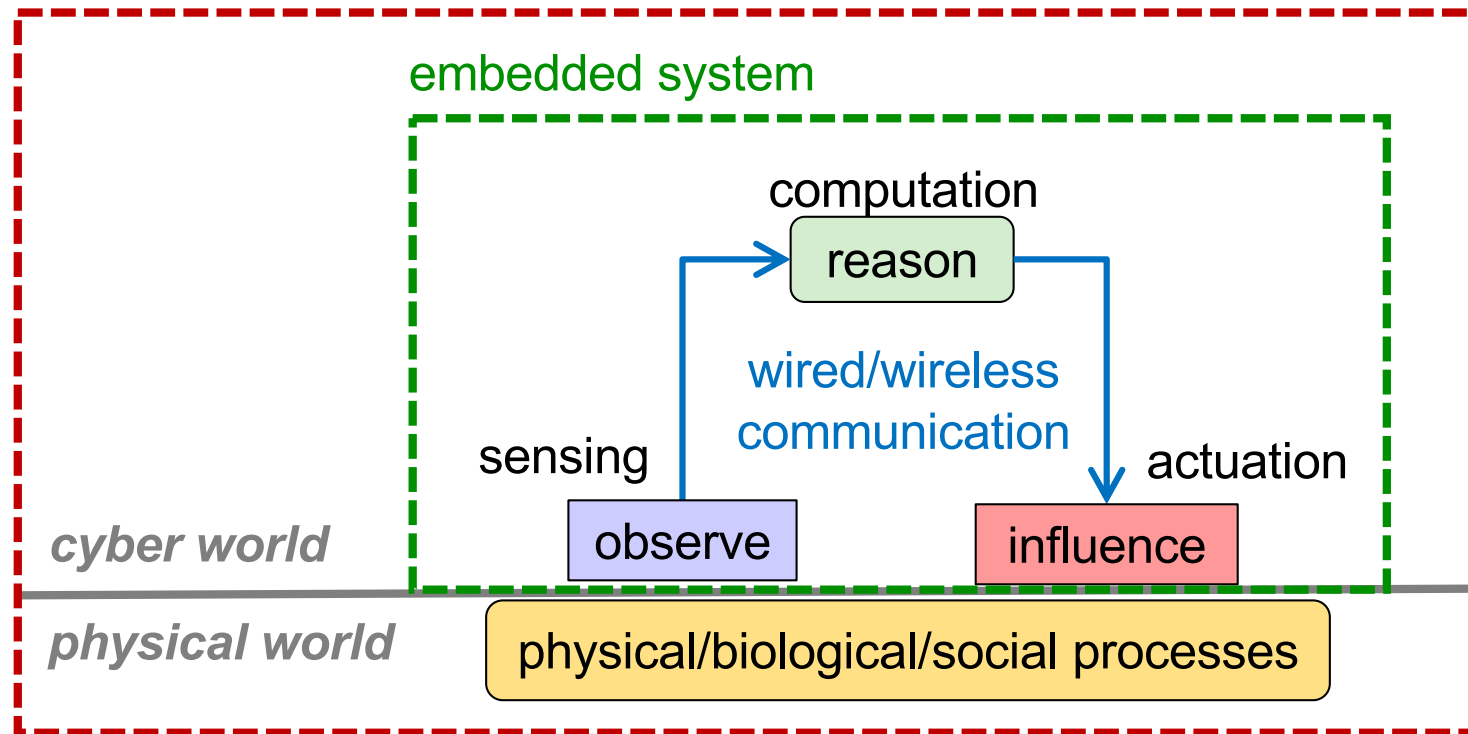
Industrial Internet of Things (IIoT)

- ... or Cyber-Physical Systems (CPS), Industry 4.0
- Digitalization as “*convergence of Operation Technology with Information Technology*”
 - Think about the revolution brought by digital into information systems, and scale it to production processes!
- Often the focus: condition monitoring and predictive maintenance
 - knowing when a part will malfunction before it actually does so



Embedded Systems & Cyber-physical Systems (CPS)

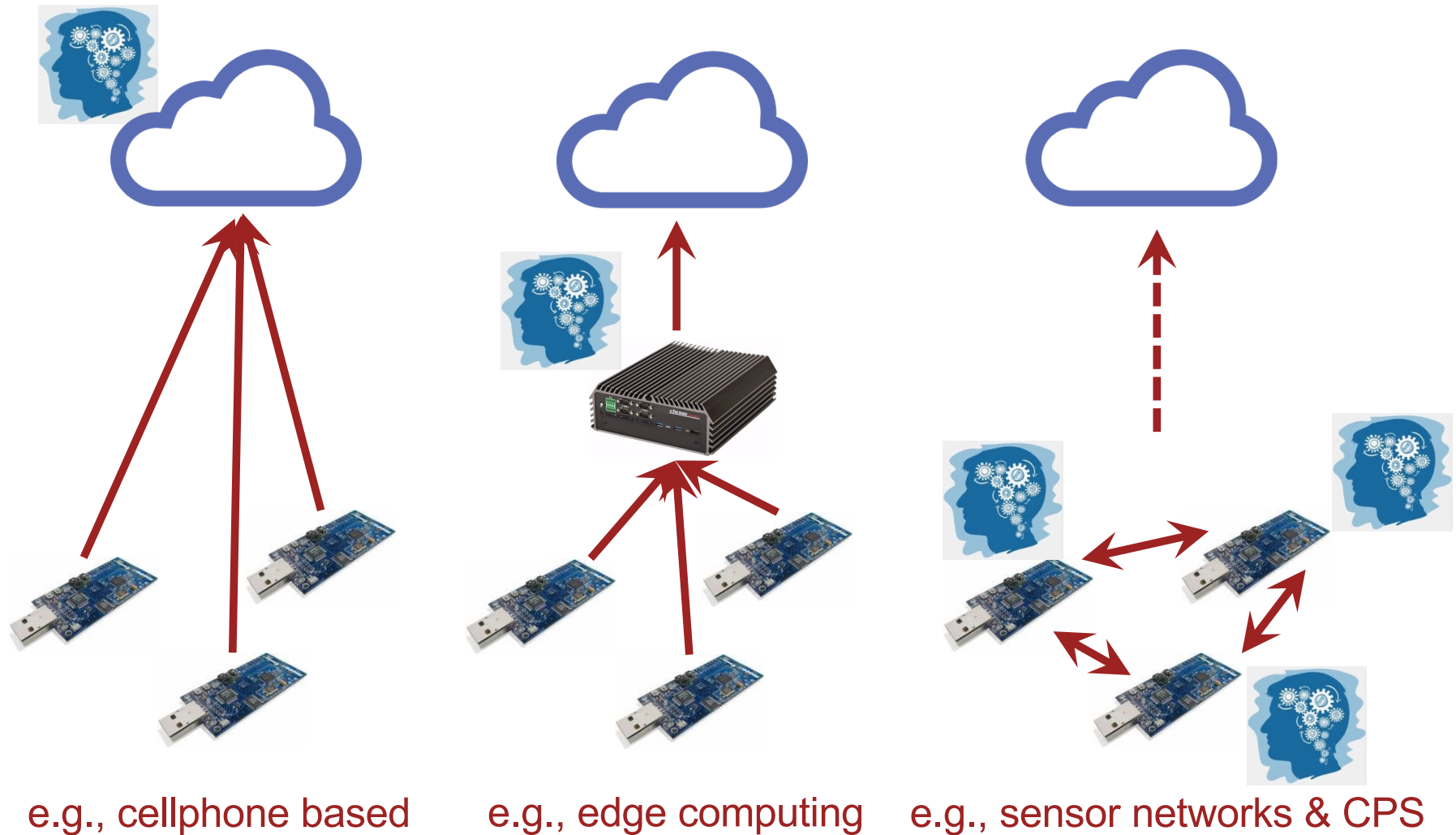
cyber-physical system (CPS)



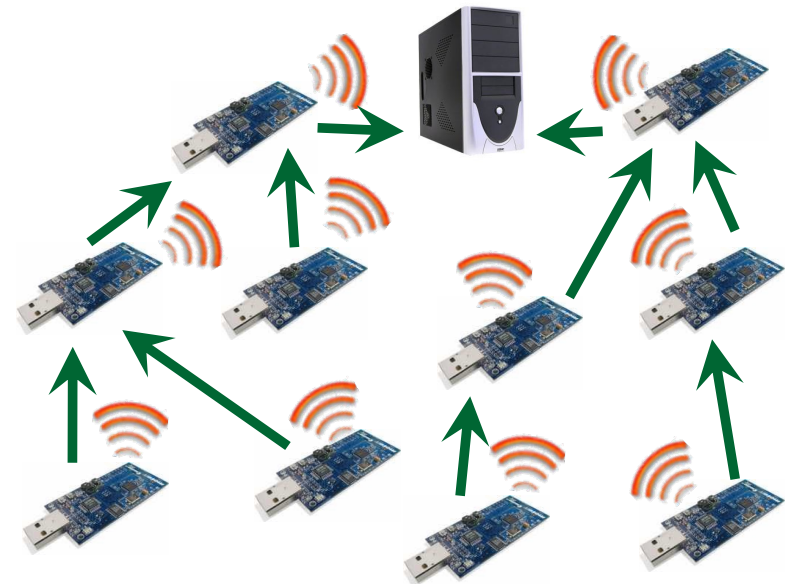
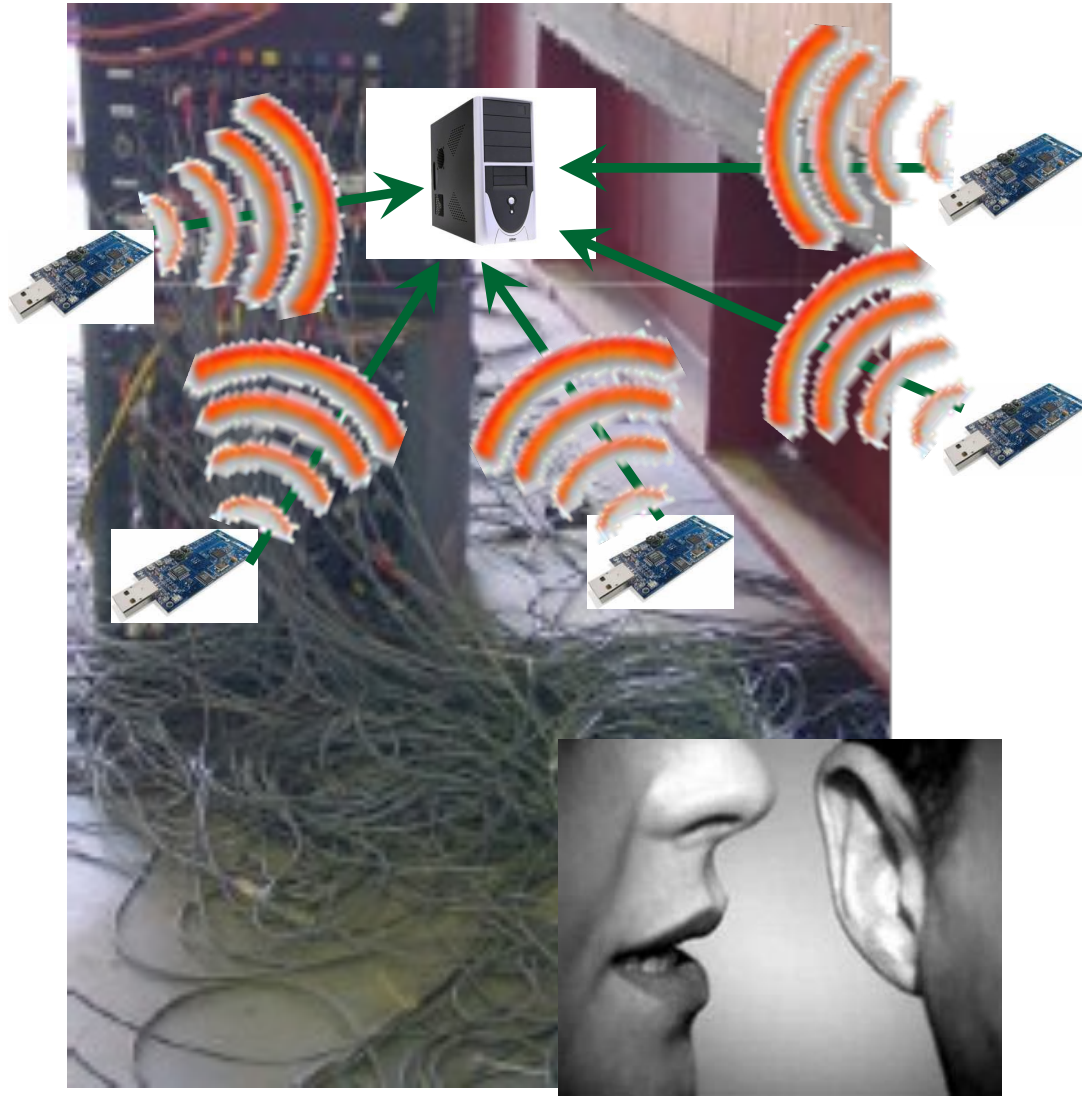
Use feedback to influence the dynamics of the physical world by taking smart decisions in the cyber world

Architecture(s) of IoT

Several Flavors of Networking ...

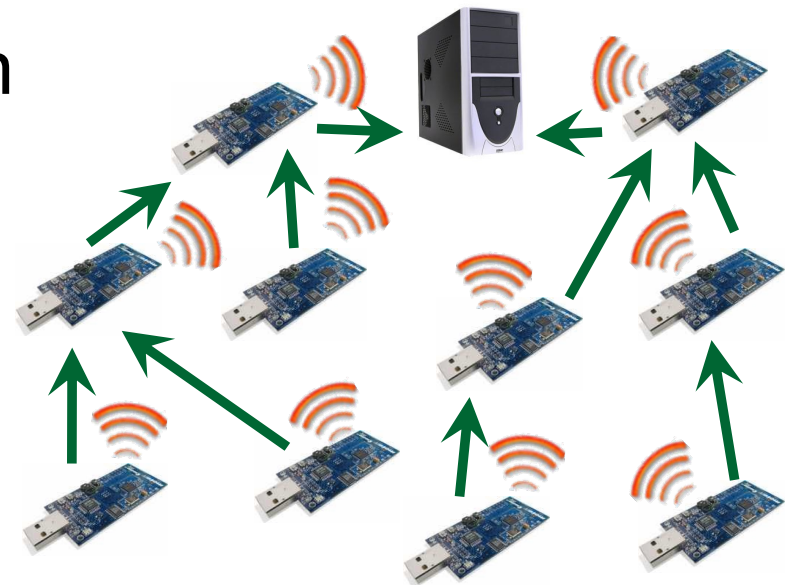
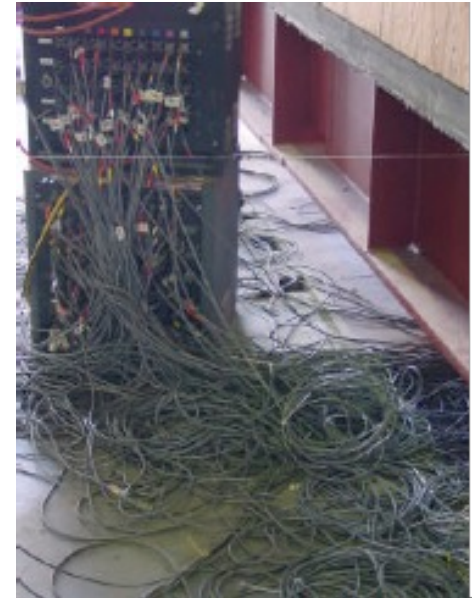


Wireless sensor networks



Wireless Sensor Networks

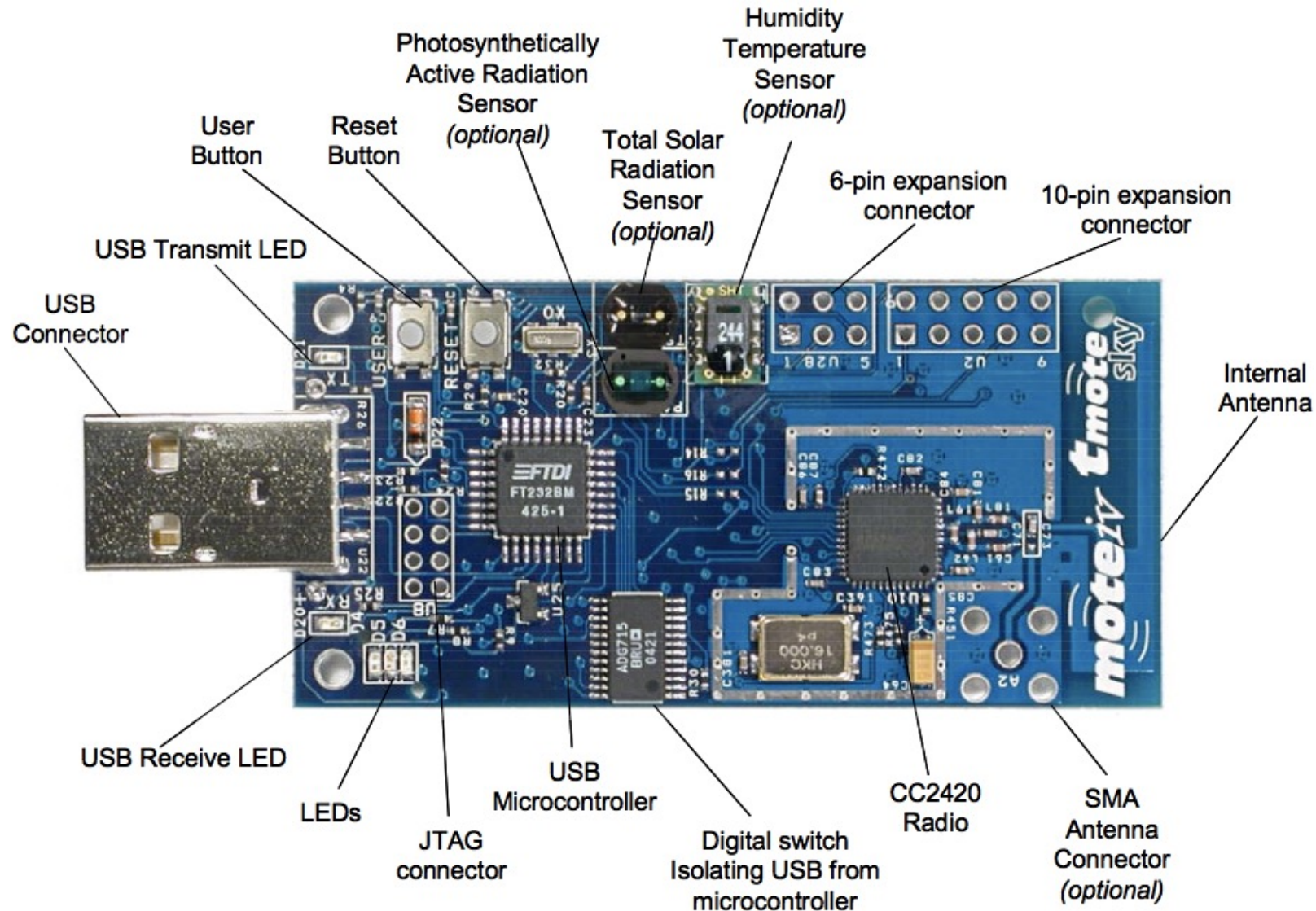
- Key component in IoT scenarios
- Replace few, accurate, expensive sensors with many cheap ones
 - coverage, flexibility, cost
- Distinctive feature:
self-organizing topology with
multi-hop communication
 - transmit power $\approx$ distance²
 - many cheap devices with short-range communication
 - more coverage with less energy (and no wires!)



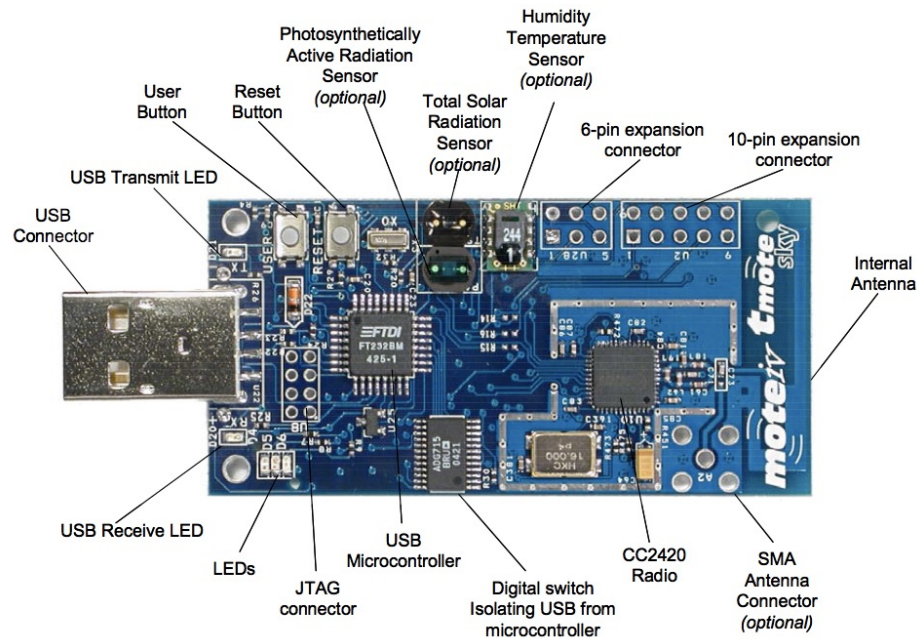
IoT, WSNs, Distributed Systems

- IoT and WSNs are examples of massively decentralized distributed systems
- They are *networked embedded systems*: they pose additional challenges
- In general, one cannot simply ignore
 - the underlying physical layer (hardware & comms)
 - the role of the deployment environment

A Classic Platform: TMote Sky



A Classic Platform: TMote Sky

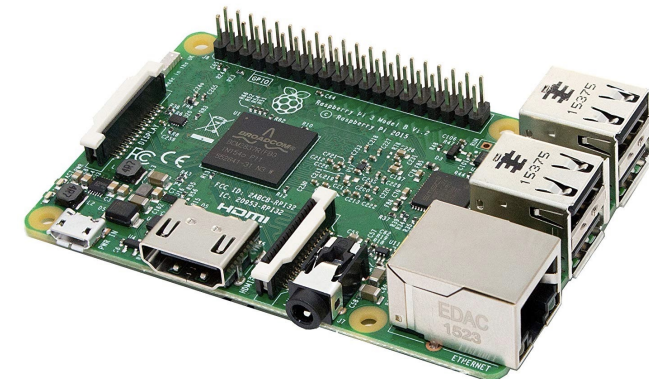


- MCU: TI MSP 430 (16 bit RISC)
 - ❑ 8 MHz
 - ❑ 10 KB RAM
 - ❑ 48 KB code, 1 MB flash
 - ❑ **Current consumption:**
~100 μ A (active), <1 μ A (idle)

In comparison:

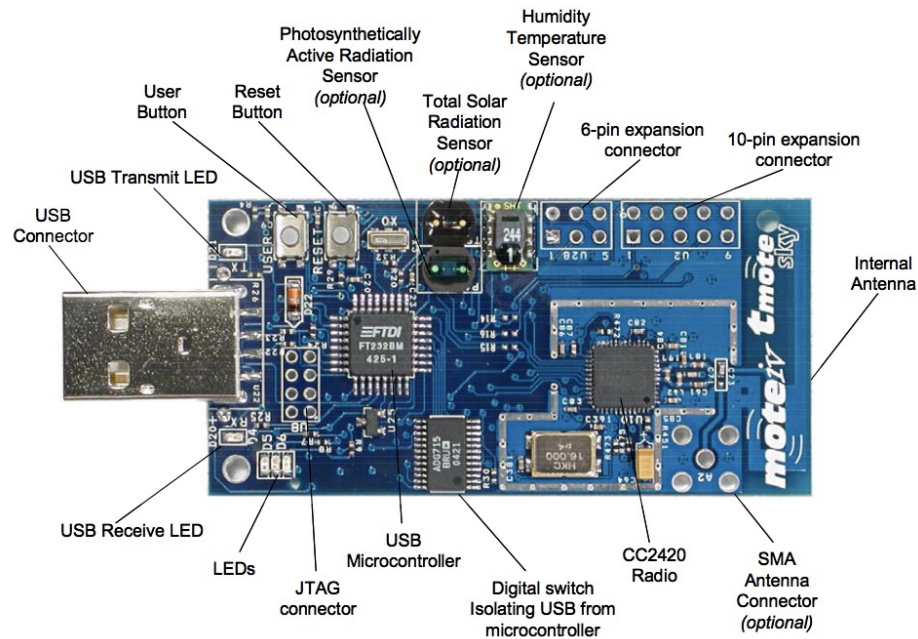


Arduino Uno: ~50 mA



Raspberry Pi: ~250 mA

A Classic Platform: TMote Sky



■ Radio: TI CC2420

- Fully IEEE 802.15.4 compliant

mode	current	power consumption
RX	18.8 mA	33.84 mW
TX	17.4 mA	31.32 mW
idle	0.426 mA	0.77 mW
sleep	0.02 mA	0.036 mW

- In comparison: WiFi and cellular phones consume in the order of Watts!

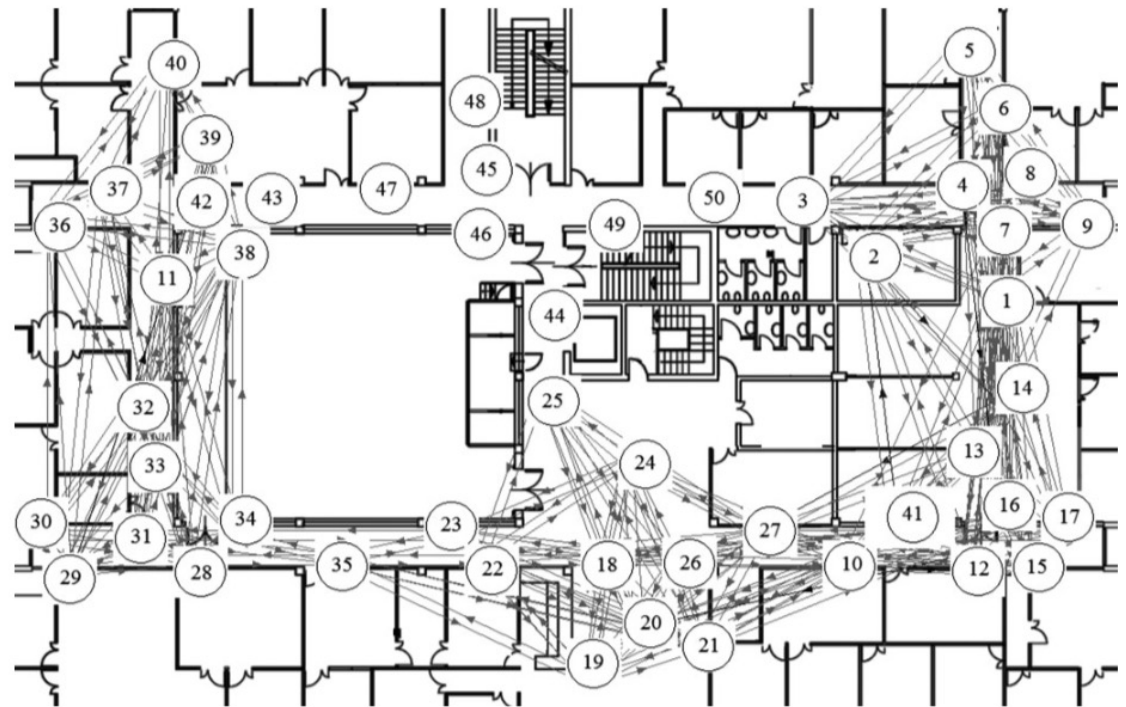
Challenges



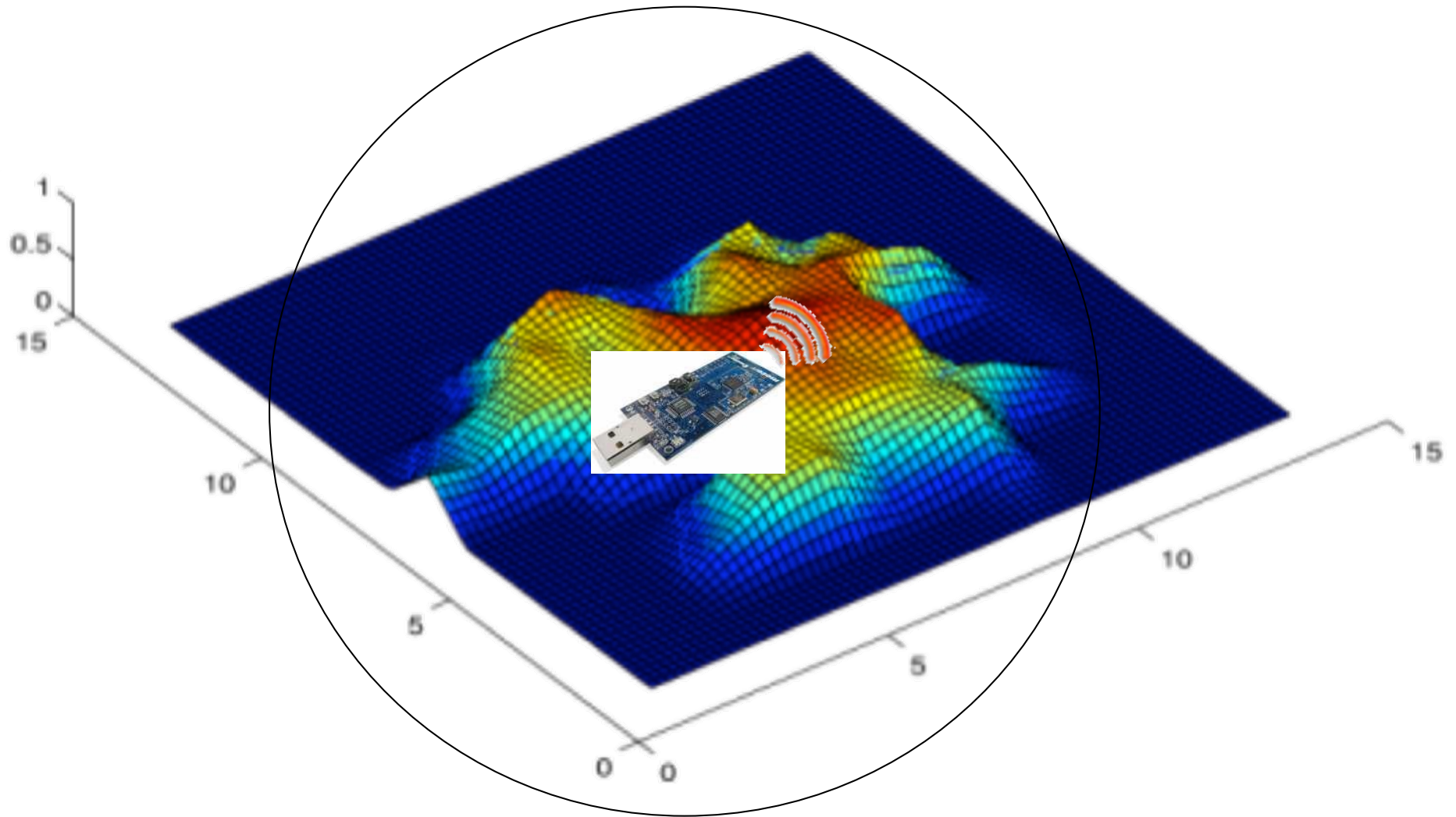
Why is Communication a Challenge?

1. The range is limited: requires multi-hop
2. The radio dominates power consumption
3. The low-power links are unstable
 - In space, time, frequency and direction!

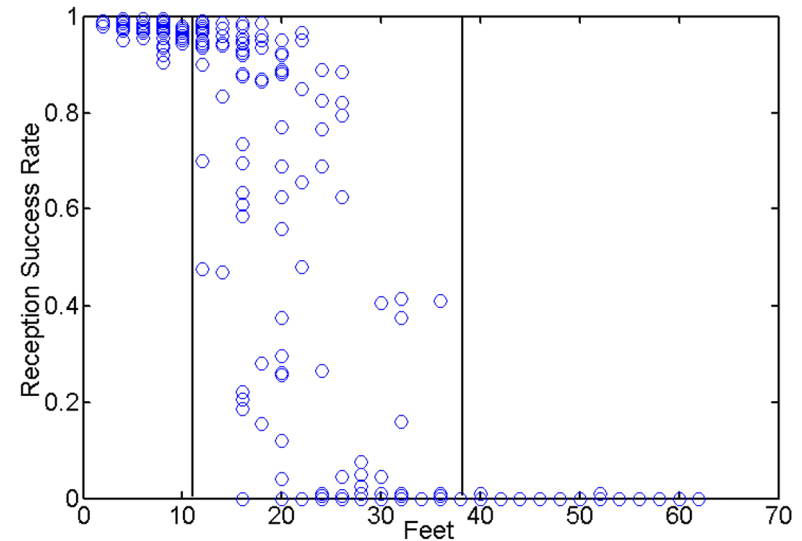
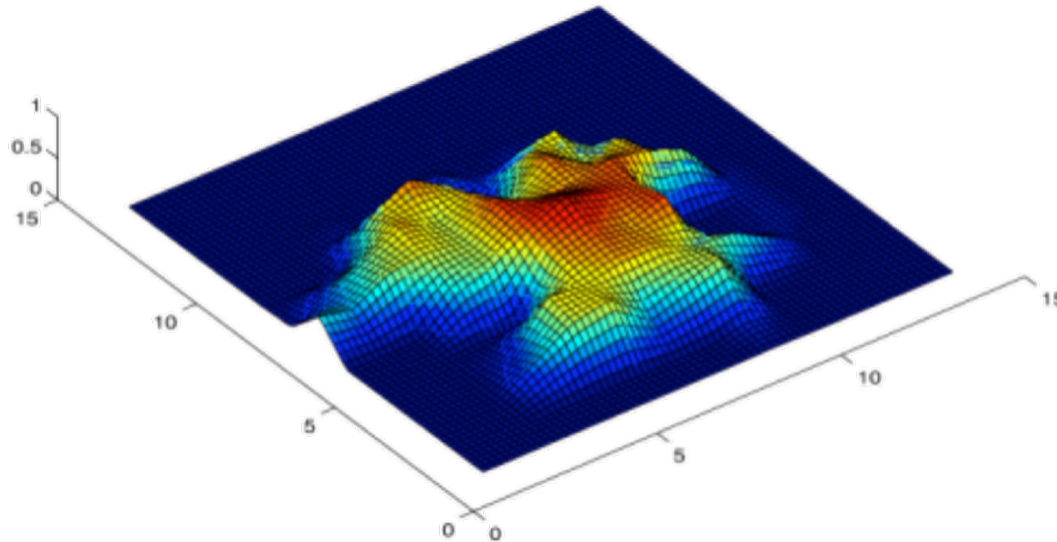
One of the testbeds
of our research group



Nice to avoid wires, but...

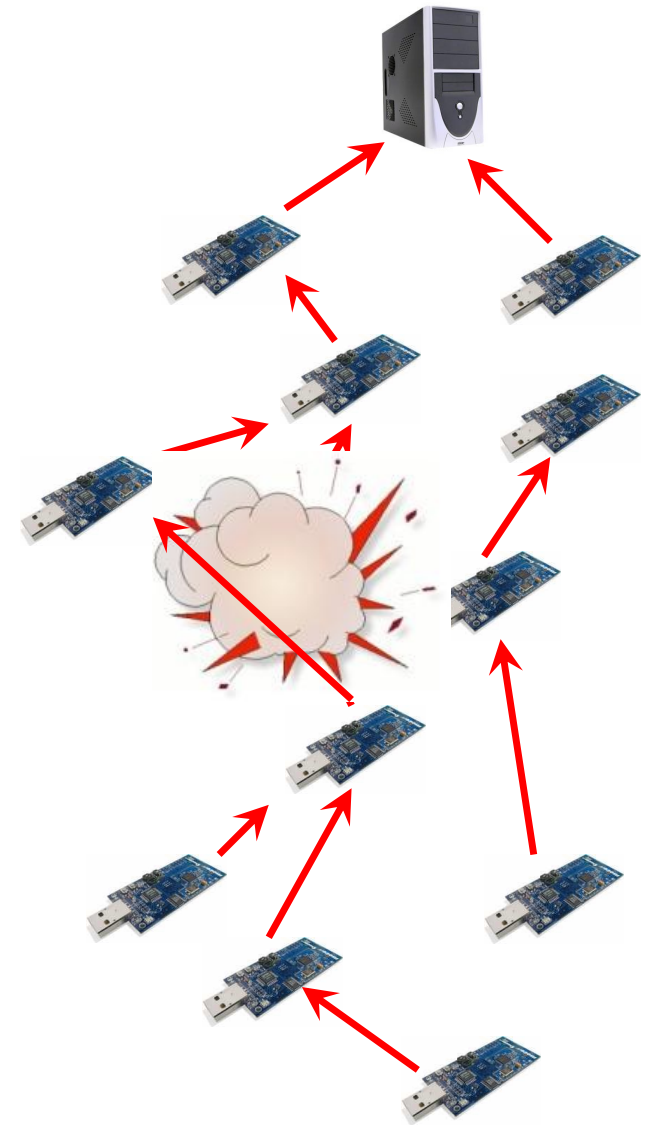
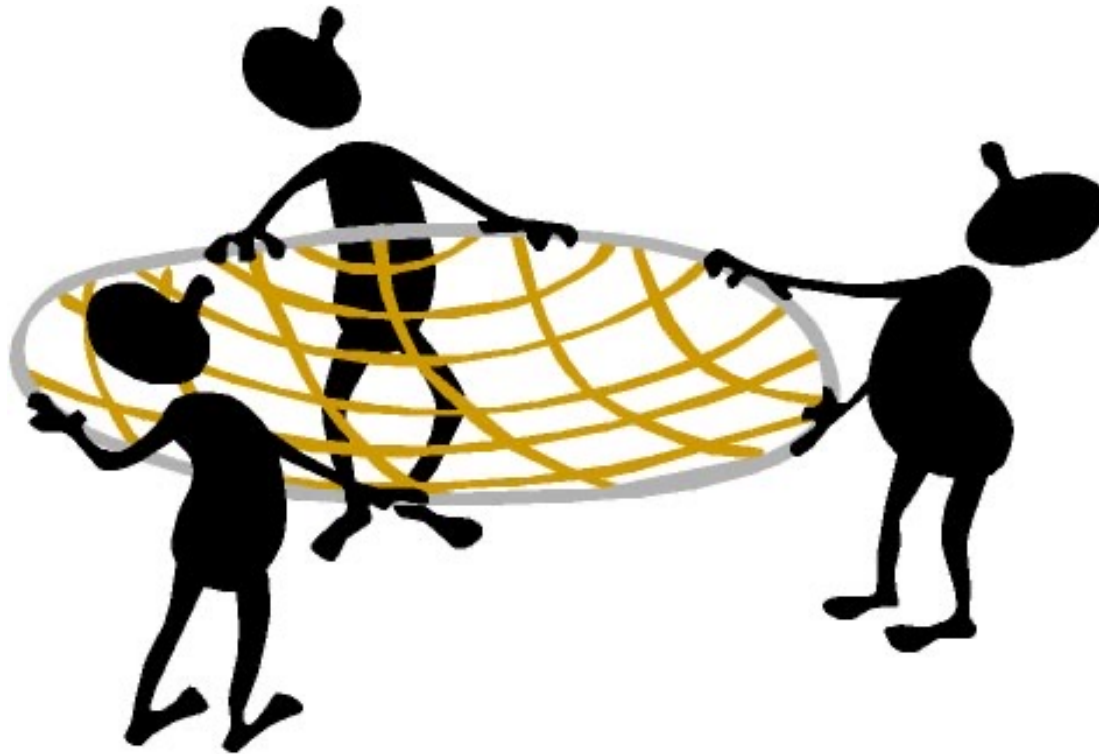


The Woes of Wireless ...



- Non-isotropic range, asymmetric links
- Collisions, hidden terminal problem
- Link quality easily affected by environmental parameters
 - e.g., temperature, humidity, presence of people
- “What-you-see-is-not-what-you-get” topologies
 - e.g., good connection to far nodes and bad to close ones

Reliability



A Few Caveats...

Communication is generally more expensive than computation

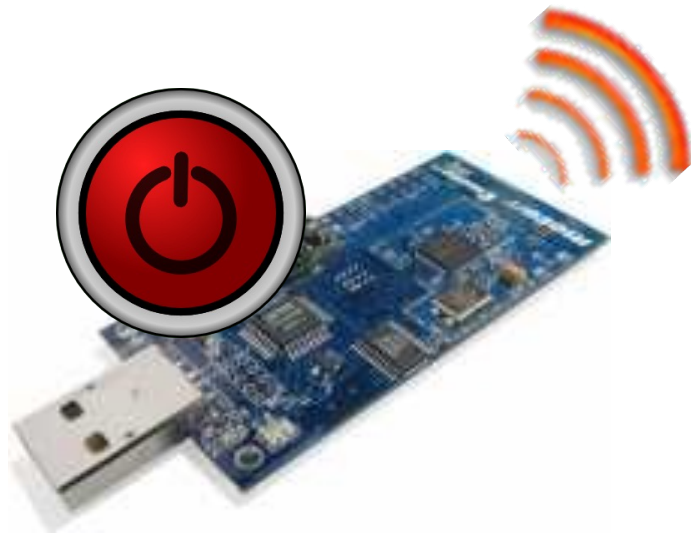
Communication requires both TX and RX!



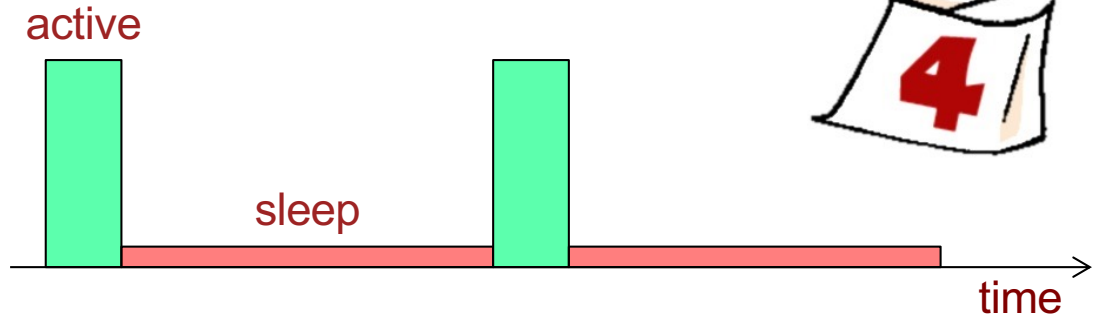
```
End Sub  
End Sub  
Private Sub tbToolBar_ButtonClicked  
On Error Resume Next  
timTimer.Enabled = True  
Select Case Button.Key  
Case "Back"  
    brwWebBrowser.GoBack  
Case "Forward"  
    brwWebBrowser.GoForward  
Case "Refresh"  
    brwWebBrowser.Refresh  
Case "Home"  
    brwWebBrowser.Home  
End Select  
End Sub
```

1 message = ~1000 instructions
(rule of thumb)

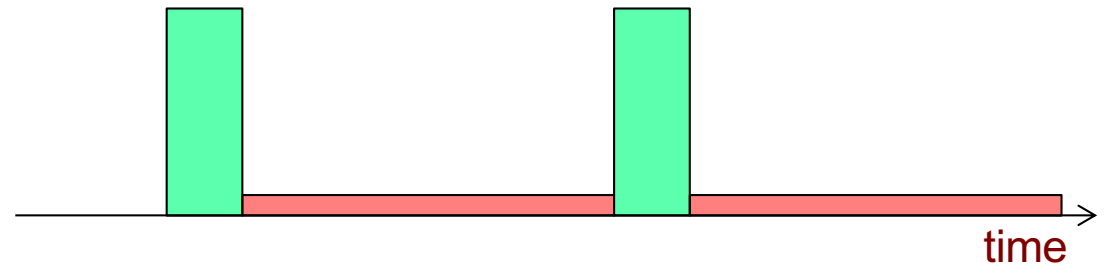
What About a Nap?



radio on: $\sim \text{mA}$
radio sleep: $\sim \mu\text{A}$

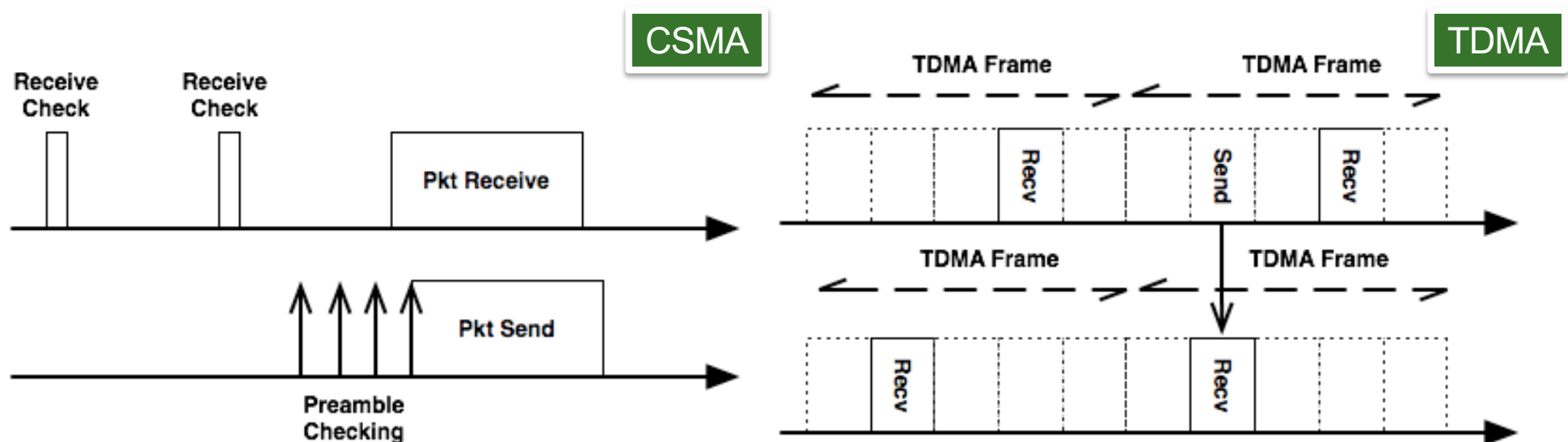


Duty cycling is a must, but brings complexity in the protocol stack



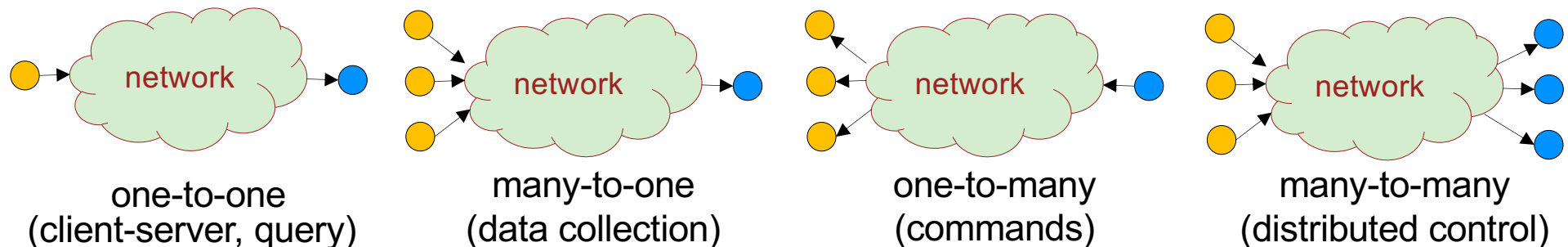
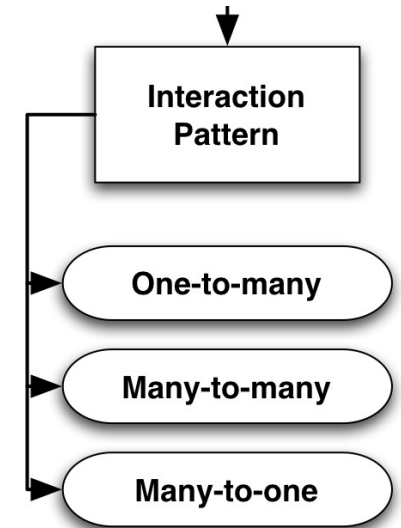
Medium Access Control (MAC)

- Only goals: avoid packet collisions and turn off the radio whenever possible
- Much simpler than conventional MACs
 - some features are demanded to application code
- CSMA (Carrier Sense) with preamble sampling and TDMA (Time Division) are dominant
 - CSMA good for changing topologies, may suffer under heavy load
 - TDMA guarantees predictable performance, needs time synch



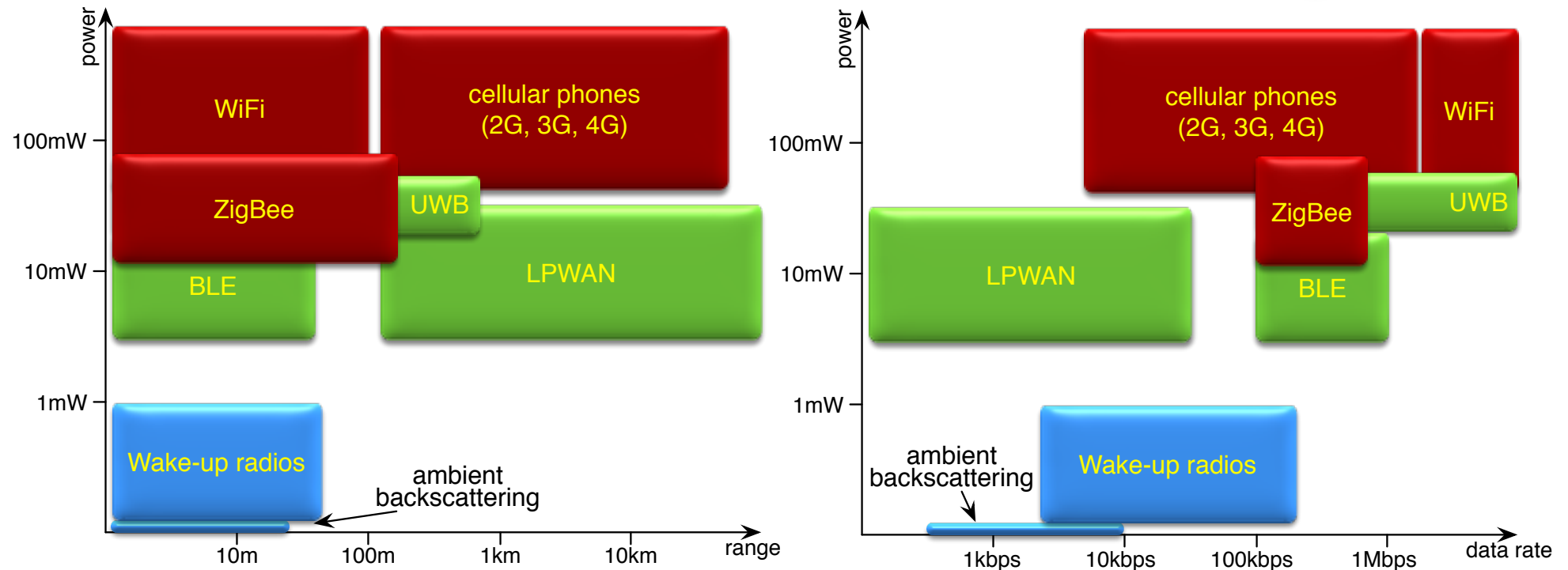
Interaction Pattern (Routing)

- Today's mainstream distributed computing is dominated by one-to-one interactions
 - e.g., client-server, based on unicast communication
- Sensing involves **many-to-one** interactions
 - many sensing sources funnel data to a single sink
 - Also called **convergecast**
- Control involves **one-to-many** interactions
 - e.g., to query for specific information, change sensing parameters, operate a set of actuators, or disseminate code updates
- **Many-to-many** interactions are becoming common
 - especially in sense-and-react applications: e.g., multiple actuators gather data from overlapping sets of sensors



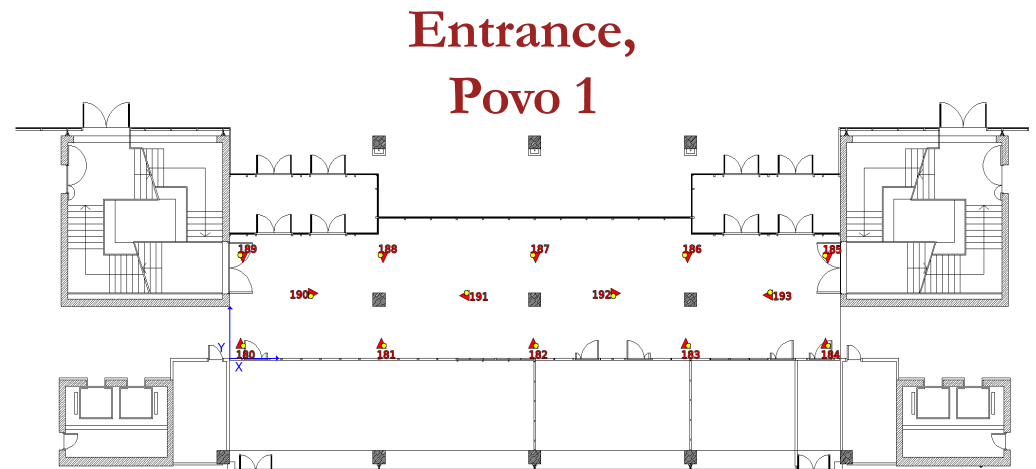
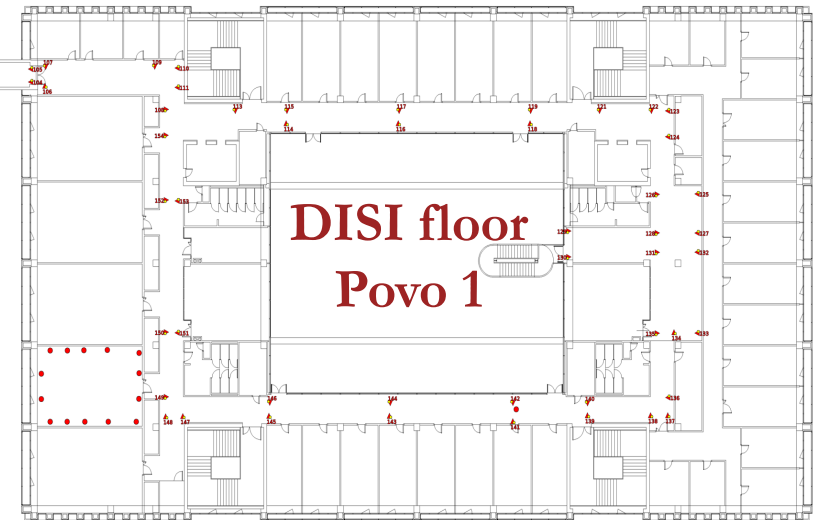
Multiple Radios

And don't forget 5G!



- Each technology strikes different tradeoffs, catering for different application scenarios in IoT
- Hardware vendors offer boards and chips equipped with multiple radios!

IoT Testbed



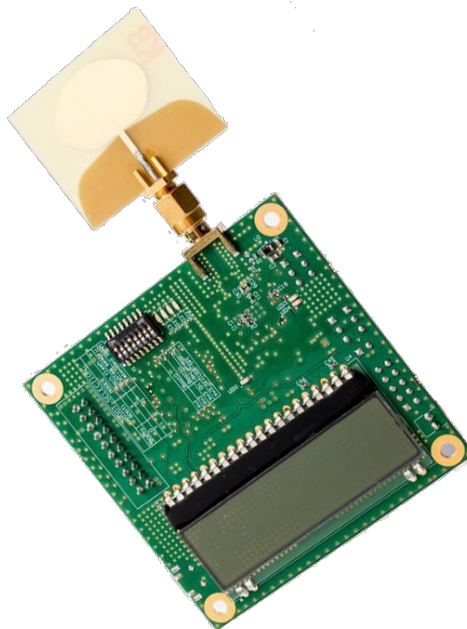
3 radio technologies, **~130** nodes, **>8000** m²

IoT Testbed: Nodes

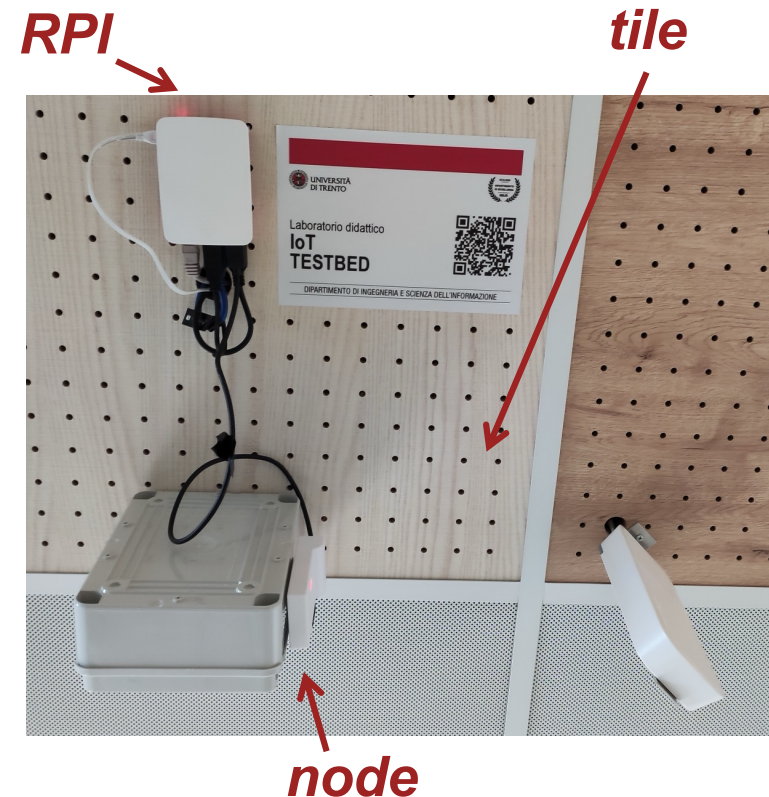
- Nodes are placed on the ceiling tiles, with power and Ethernet cables above it
- Each node is paired with a Raspberry PI 3
- Users can **remotely** reprogram nodes, control experiments, and gather data



Zolertia Firefly
IEEE 802.15.4
narrowband



Decawave EVB1000
IEEE 802.15.4
ultra-wideband (UWB)

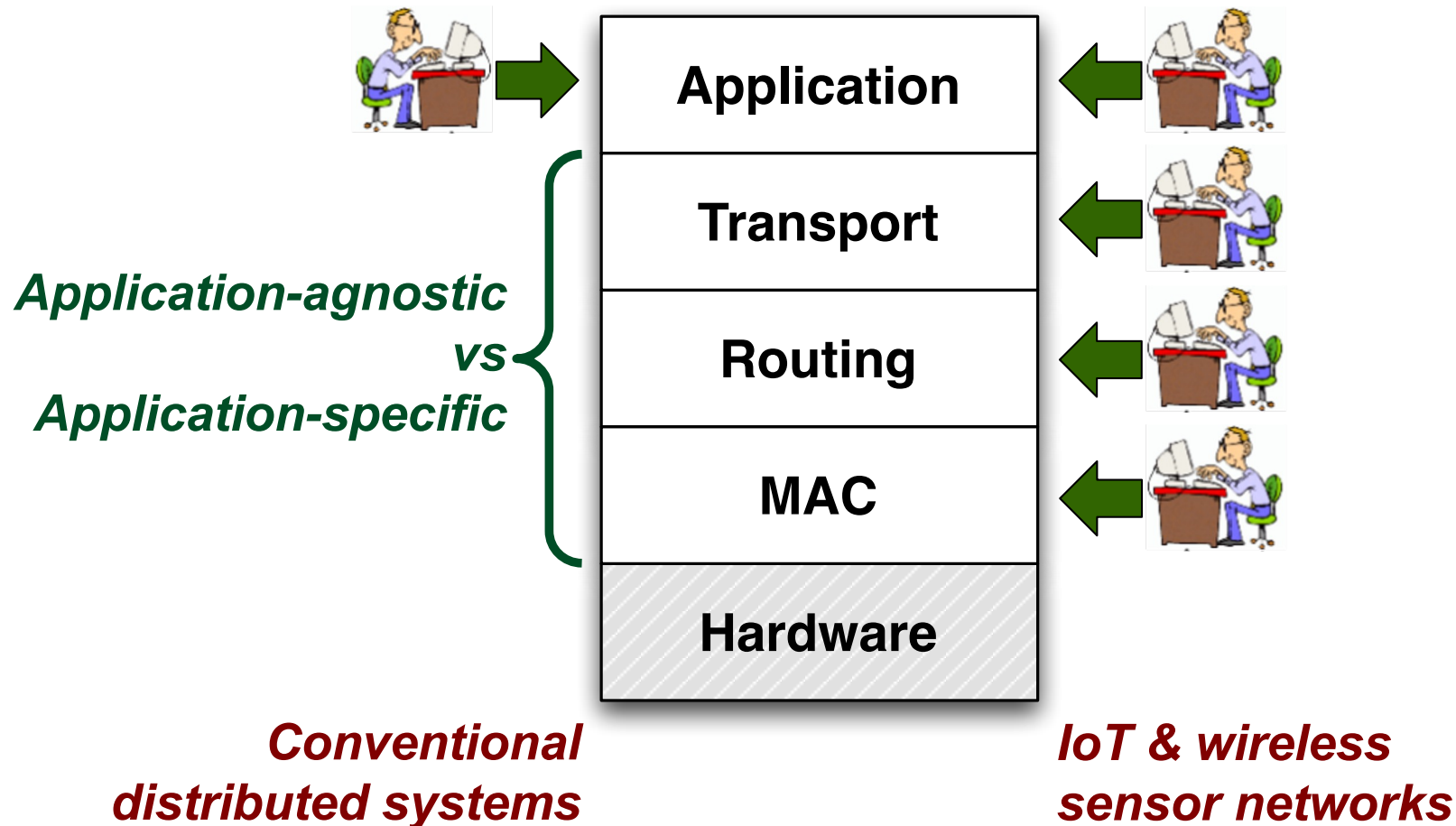


- Each node hosts two types of devices, common for:
 - wireless sensor networks
 - ranging and localization

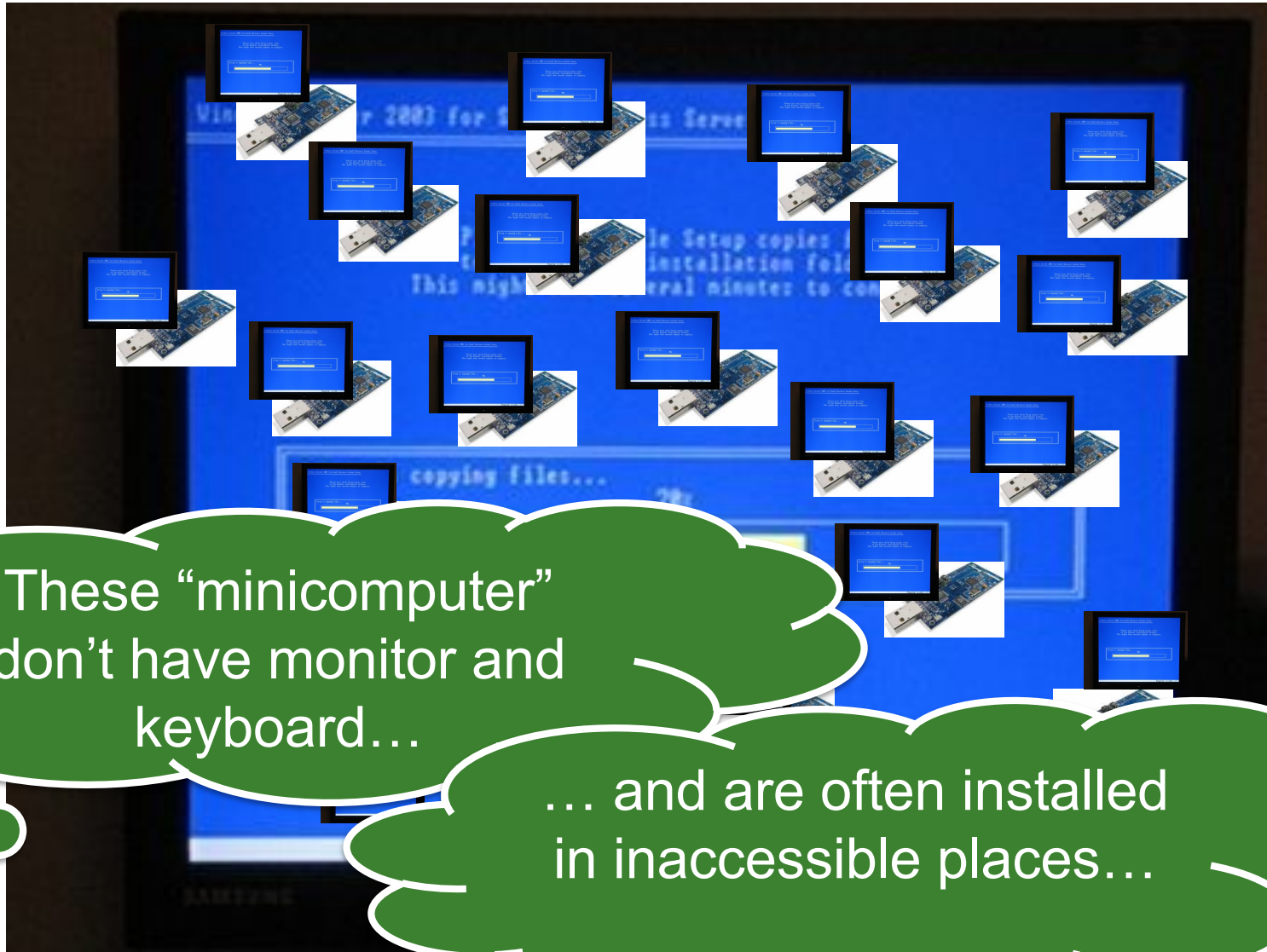
Operating Systems for WSNs

- Different from mainstream ones
 - basic run-time support for application programs
 - no support for user interaction
- Programs are **cross-compiled** and linked with the OS library: the resulting binary is deployed on the WSN node
- Contiki is a popular choice today
 - used by many state-of-the-art systems
 - ported on an increasing number of platforms
- Many others exist
 - TinyOS (old), RIOT, FreeRTOS, Mynewt, Zephyr...

The Role of Software in WSNs



Reprogramming

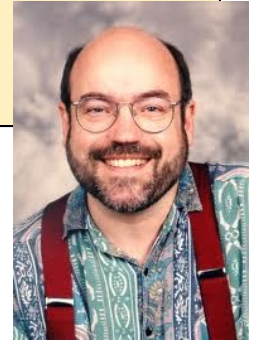


These “minicomputer”
don’t have monitor and
keyboard...

... and are often installed
in inaccessible places...

Conclusions

The most profound technologies are those that disappear. They weave themselves into the fabric of everyday life until they are indistinguishable from it.



- It's not just a slogan
- From research to everyday reality
 - ... they are already here!
- Impact on the individual and society
- Require dedicated techniques and a different, cross-layer mindset

“Low-power Wireless Networking for the Internet of Things”
course on 1st semester